MAPPING TREE-LEVEL STRUCTURAL DIVERSITY FOR PRECISION HARVESTING OPERATIONS:

A COMPARISON OF 5-CM DRONE AND 50-CM AERIAL DATA

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THE IMPORTANCE OF FOREST STRUCTURE ANALYSIS FOR FOREST OPERATION

Tree-level forest structural diversity

- Reflect local variations of forest structural diversity
- Indicates areas with high tree density
- Highlights regions with large variations in tree height and species composition

Harvesting operation

Assess stand stability, guides thinning or restoration practices

MAPPING TREE-LEVEL STRUCTURAL DIVERSITY FOR PRECISION HARVESTING OPERATIONS

- Individual tree detection using multi-source data
- Tree-level forest structural diversity mapping
- Clustering to support practical forest operations

MAPPING TREE-LEVEL STRUCTURAL DIVERSITY FOR PRECISION HARVESTING OPERATIONS

Ground truth for evaluation

Field data + Drone data → ITD_{GT}

Deep learning-based ITD

Drone data

Aerial data

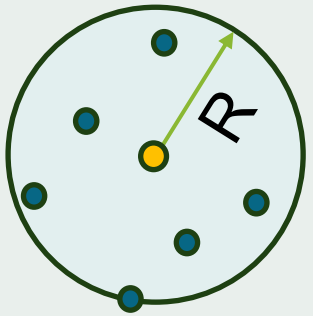
DLS ALS → CHM

Spectral: R, G, B, NIR

ITD_{drone}

ITD_{aerial}

Spatial modeling



Structural diversity indices

- Spatial tree density
- SMLI of tree species
- SGINI of tree height

Forest structural indices mapping

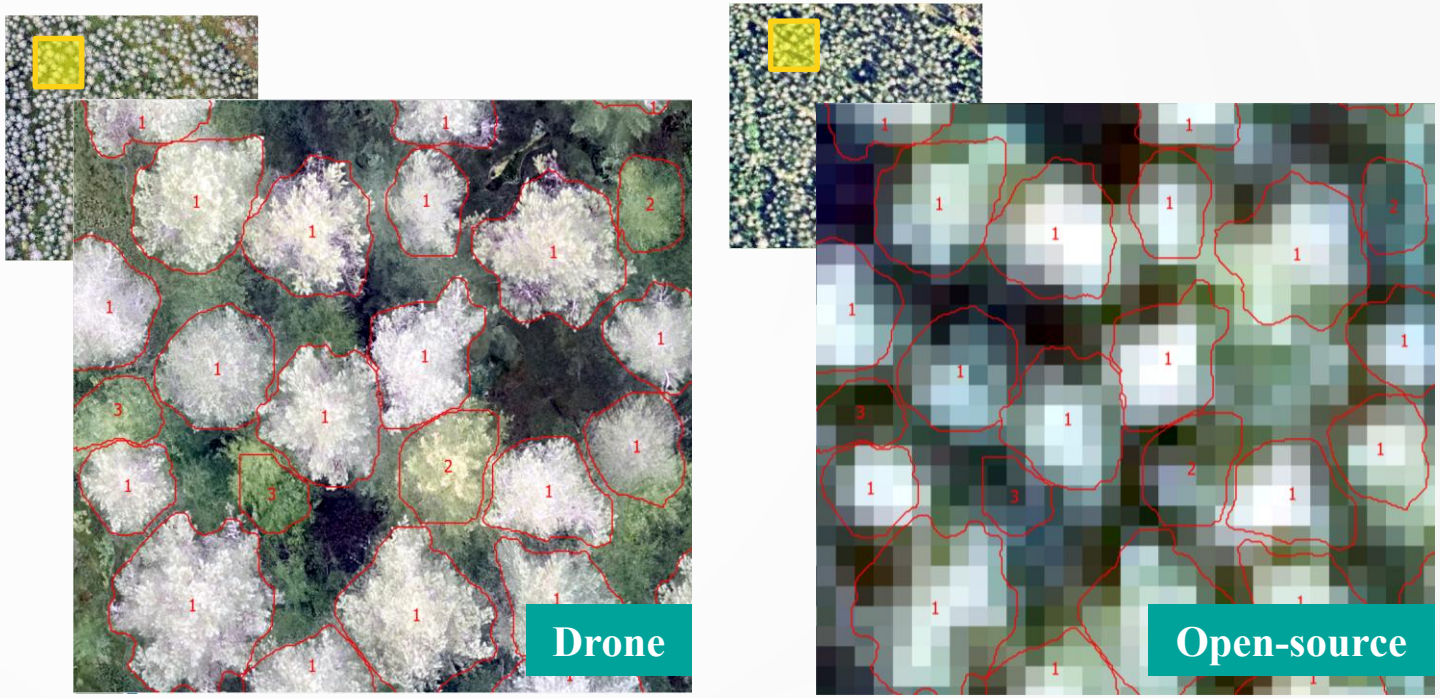
Drone VS. Aerail

Clustering to support practical forest operations

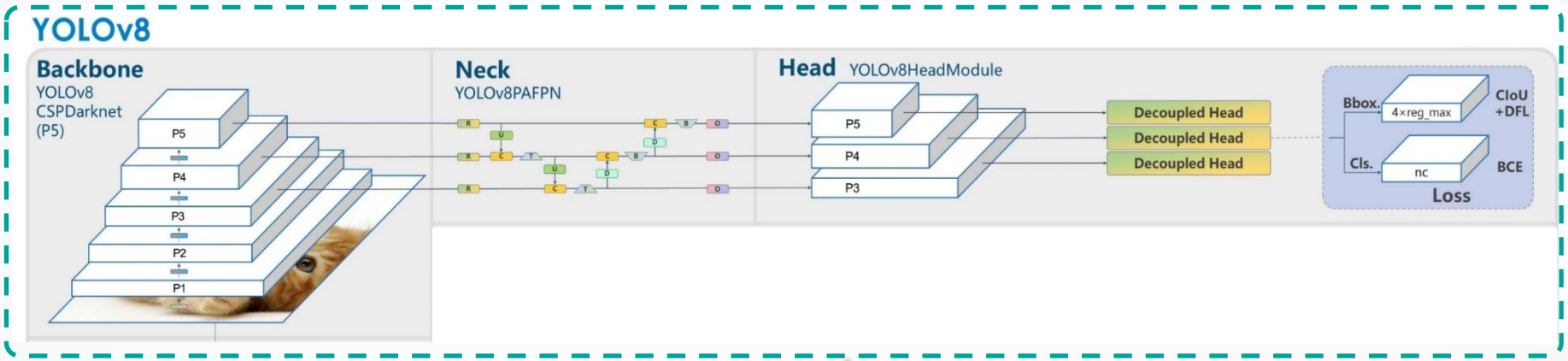
INDIVIDUAL TREE DETECTION FROM REMOTE SENSING DATA

Individual tree detection

Detecting tree species, crown boundary, and position using remote sensing and deep learning-based instance segmentation methods



WORKFLOW OF INDIVIDUAL TREE DETECTION



Input data

Lidar data:
CHM

Spectral:
R, G, B, NIR

Drone: 5 cm

Open-source:
50 cm

deep learning-based instance segmentation

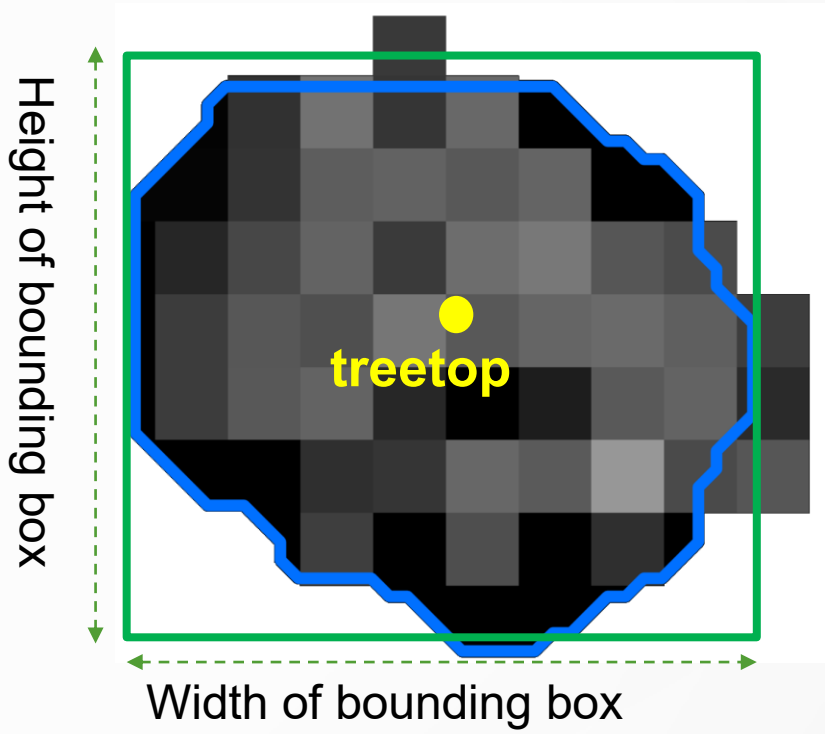
Super-resolution
Upscale X 4

Output

- Tree location
- Species
- crown boundary

CALCULATE TREE HEIGHT FROM ITD RESULTS

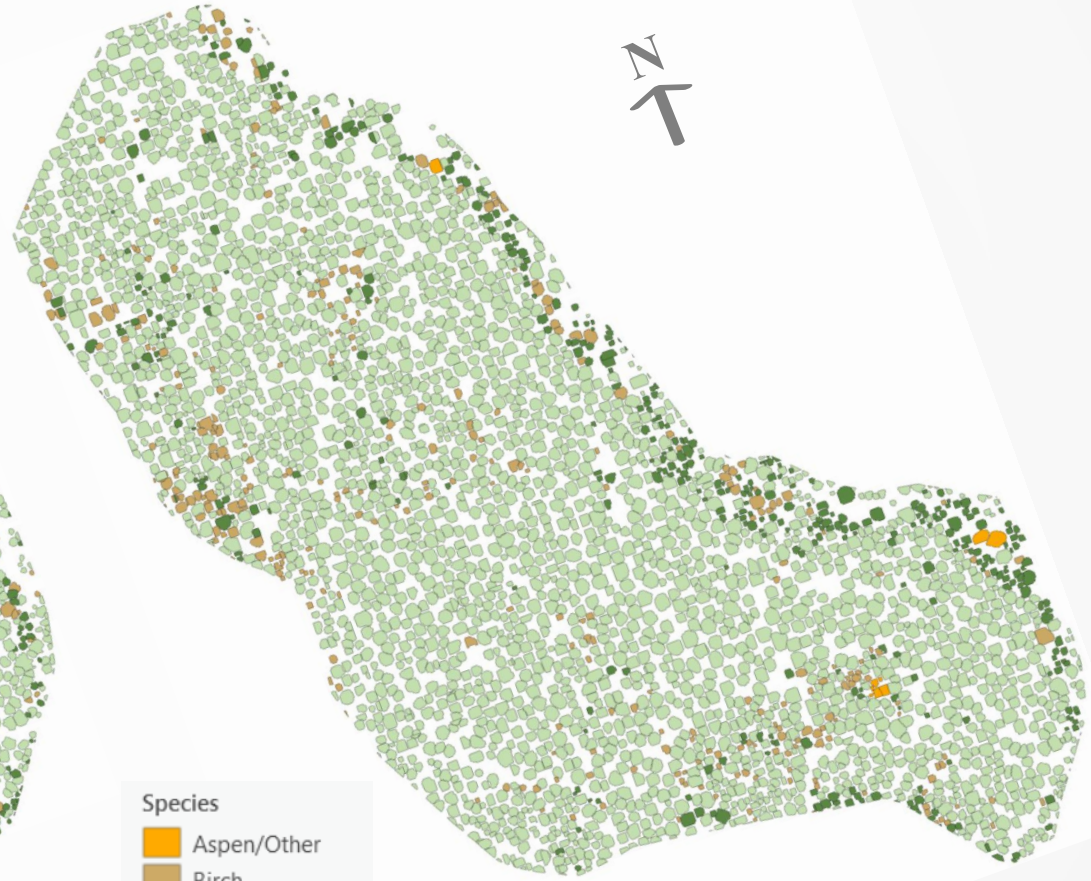
- **Tree center**: Detected as the local maximum within the predicted tree boundary.
- **Tree height**: the CHM value at the detected **treetop.**



ITD DRONE VS. AERIAL DATA

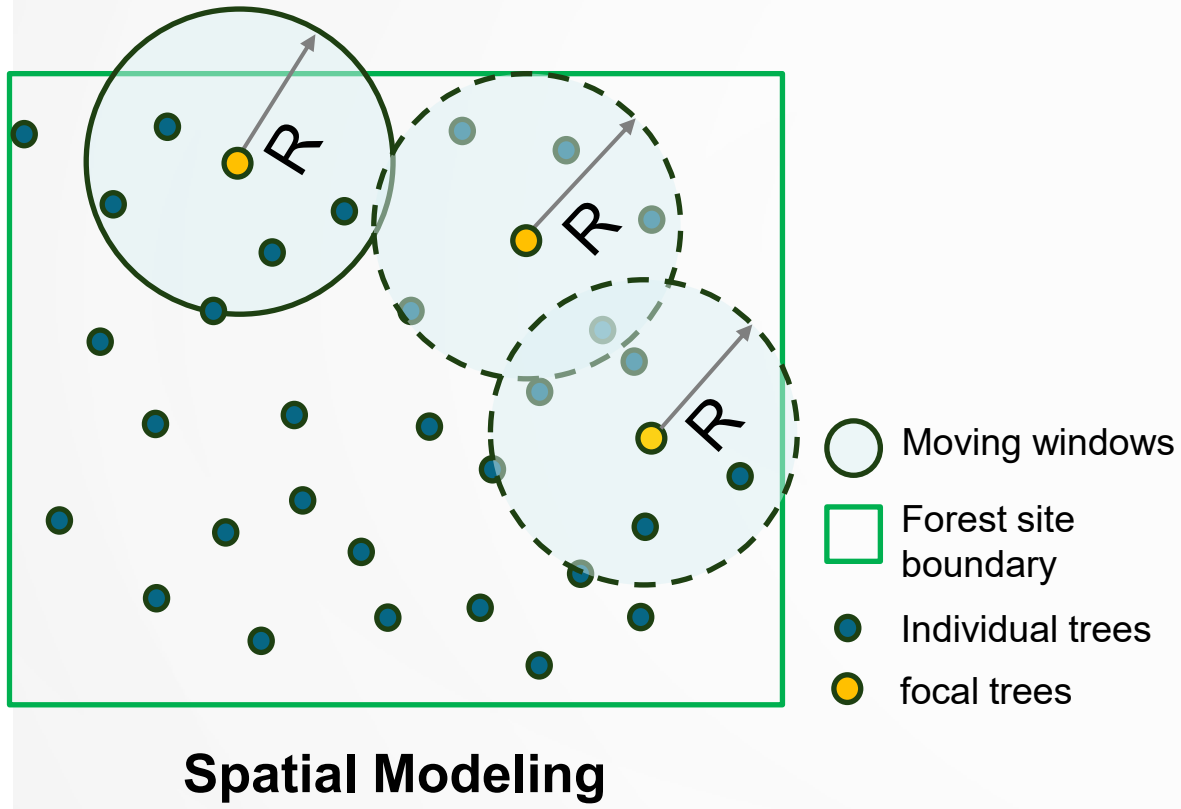


Drone



Open-source

TREE-LEVEL FOREST STRUCTURAL INDICES



Spatial Modeling

- **Tree density**

the number of trees per unit area. scaled up to a hectare for standardization

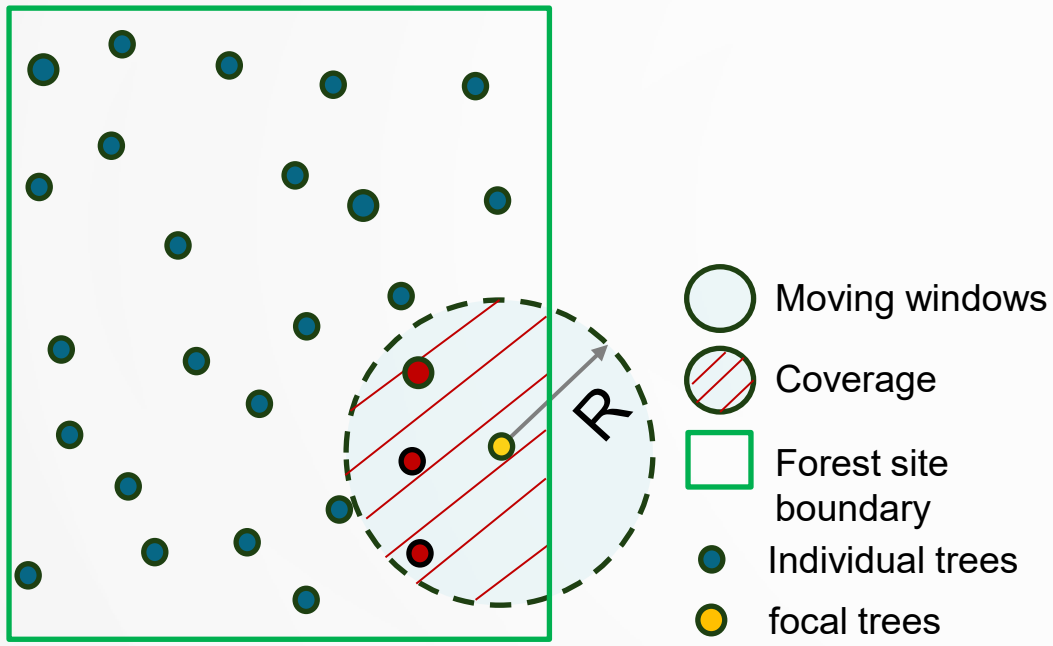
- **Spatial GINI index of tree height**

Measures the unevenness of tree height.
Low GINI (≈ 0): similar heights, a uniform vertical structure.
High GINI (≈ 1): trees differ greatly in height.

- **Spatial Mingling index of tree species**

Evaluates the diversity of tree species
Low MLI (≈ 0): Tree species are fairly equal.
High MLI (≈ 1): Tree species are highly unequal.

EDGE EFFECT CORRECTION



Edge effect correction

$$Covergae\ ratio = \frac{area\ of\ (circle\ \cap site\ boundary)}{area\ of\ circle}$$

- Coverage ratio < 1: the focal tree near the edge
- Consider the trees only inside the forest boundary

CALCULATE FOREST STRUCTURAL INDICES

RS data ⇨ Tree location, species, and crown boundary ⇨ Tree height ⇨ Density, SGINI, SMLI R=10 meters



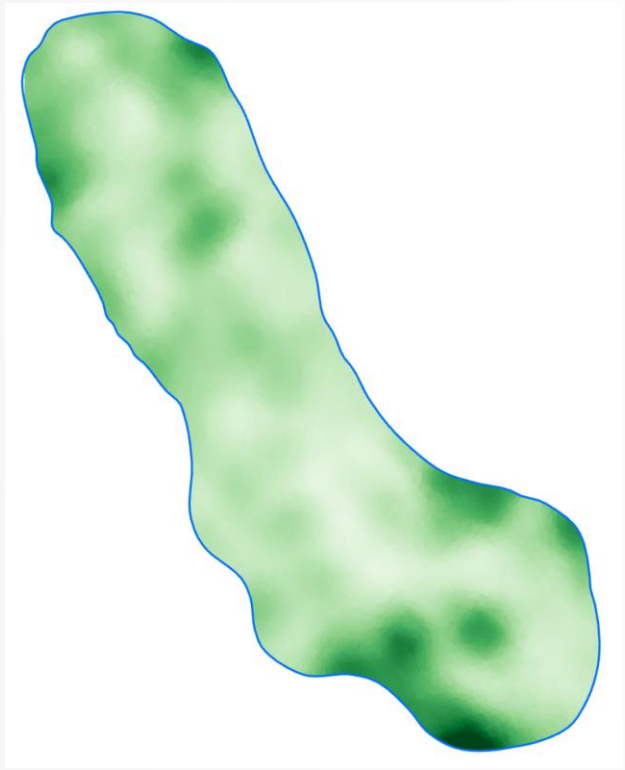
Class	Confidence	KRK_NIMI	Species	Crown_leng	height	Shape_Area	Vdm3	DBH	Clr_R	Ins_r	Diameter	radius	Tnum	Tnum_ha	Deciduous	Mheight	Mcrown	GINI	Edge_ratio	MLISp
1	87.149233	Jokisalo	Pine	7.68687	9.927154	3.9044	65.54212	126.1	1.571388				9	99			5.91			
2	42.007786	Jokisalo	Spruce	4.920813	5.928769	1.6522	29.472418	109.1	1.025808				0	107			5.449			17
2	84.409124	Jokisalo	Spruce	5.919002	3.494923	2.3743	22.155511	112.9	1.189287				2	96			4.887			75
3	62.165022	Jokisalo	Birch	12.643428	10.362538	9.2469	89.683757	153.9	2.549862	1.54984	4.099702	2.024794	23	1093.633905	15	8.288	5.195	0.209	0.670508	0.454545
1	47.606638	Jokisalo	Pine	5.844981	9.313769	0.9898	66.601045	130.6	1.008918	0.67493	1.683848	0.749923	19	1118.927488	11	7.893	4.764	0.215	0.541377	0.777778
2	63.255751	Jokisalo	Spruce	8.051954	6.622461	2.3161	43.472589	128.1	1.502551	1.049892	2.552443	1.074889	11	868.846613	5	6.558	5.428	0.273	0.403643	0.9
2	44.419548	Jokisalo	Spruce	5.981363	3.830846	2.4507	23.603004	114.3	1.290458	0.899907	2.190365	0.924905	25	1012.811739	15	7.758	4.838	0.246	0.786972	0.875
1	41.855821	Jokisalo	Pine	5.727112	13.498693	1.7168	110.013392	143.3	1.222831	0.799917	2.022749	0.924905	10	603.7351	0	15.754	9.484	0.081	0.528082	0
3	76.255405	Jokisalo	Birch	7.106983	9.162616	2.3901	44.918538	113.8	1.326749	0.899907	2.226656	0.974899	29	980.454818	19	10.169	5.775	0.172	0.943015	0.392857
3	43.005905	Jokisalo	Birch	7.076225	6.244538	3.2443	34.387372	115.4	1.530365	0.949902	2.480267	1.199876	27	860.818829	19	10.346	6.334	0.171	1	0.307692
1	50.216991	Jokisalo	Pine	5.772806	3.262461	1.7678	22.320337	105	1.228503	0.862411	2.090914	0.87491	19	926.689605	10	6.982	4.717	0.259	0.653683	0.777778
3	81.652421	Jokisalo	Birch	12.145551	10.949538	9.1931	108.149197	165.4	2.565982	1.724822	4.290805	1.899804	30	963.880242	19	7.932	4.719	0.243	0.992307	0.413793
1	87.394142	Jokisalo	Pine	9.979479	3.429538	6.4518	31.236213	123.5	2.02483	1.387357	3.412187	1.474848	24	901.909895	12	6.982	4.505	0.281	0.848391	0.782609
2	69.792908	Jokisalo	Spruce	5.320771	3.246923	1.9896	17.810009	101.5	1.101022	0.699928	1.80095	0.849912	25	882.775437	13	7.737	4.778	0.266	0.902896	0.791667
3	84.079111	Jokisalo	Birch	5.062219	3.649	1.6047	19.297769	93.3	1.06114	0.724925	1.786065	0.77492	18	742.30235	8	5.857	3.189	0.3	0.773107	0.647059
2	89.044601	Jokisalo	Spruce	6.227785	4.812154	2.5695	38.043198	137.7	1.30849	0.899907	2.208397	0.949902	20	637.643577	2	14.425	7.694	0.199	1	0.842105
2	44.950762	Jokisalo	Spruce	6.939843	10.679615	2.9906	89.984639	147	1.42112	0.899907	2.321028	1.099887	19	605.761398	0	16.75	9.351	0.073	1	0.944444
3	85.8437	Jokisalo	Birch	15.720667	11.099384	7.3369	84.856664	144.4	2.808179	1.799814	4.607994	2.155583	30	956.465366	17	12.952	6.32	0.118	1	0.482759
2	86.497021	Jokisalo	Spruce	5.113686	3.515923	1.7671	20.252583	107.3	1.007678	0.699928	1.707606	0.724925	19	722.095046	8	5.709	3.084	0.302	0.838894	0.666667
2	49.158615	Jokisalo	Spruce	4.276019	3.048846	1.2022	16.520071	98.1	0.890133	0.549943	1.440076	0.699928	13	434.718943	6	4.933	3.382	0.286	0.953417	0.75
3	47.561914	Jokisalo	Birch	3.134524	1.986538	0.6647	15.653705	55	0.637386	0.424956	1.062342	0.475051	14	480.00261	7	4.841	3.255	0.277	0.929892	0.615385
3	43.197107	Jokisalo	Birch	9.953733	16.305386	6.2825	208.352685	188.5	2.069962	1.399856	3.469817	1.524843	20	637.643577	1	17.106	9.465	0.039	1	1
2	81.664407	Jokisalo	Spruce	6.510598	10.82877	2.5595	84.734109	140.9	1.39	0.849912	2.239913	1.099887	22	701.407935	2	14.137	7.338	0.182	1	0.809524
2	69.925743	Jokisalo	Spruce	5.827826	5.967	2.1571	41.340585	131.1	1.237814	0.849912	2.087726	0.899907	25	797.054472	2	14.444	7.523	0.163	1	0.75
3	72.7144	Jokisalo	Birch	6.581312	5.255077	2.6832	36.267463	124.5	1.361157	0.949902	2.311059	0.974902	21	669.525756	2	14.344	7.415	0.192	1	0.95
3	85.386831	Jokisalo	Birch	14.121093	13.259077	12.0025	152.436395	179.4	2.793681	1.924804	4.718485	2.024791	29	924.583187	15	9.967	4.961	0.254	1	0.535714
3	46.178433	Jokisalo	Birch	8.522523	12.665154	4.5103	90.880062	139.7	1.735476	1.149881	2.885358	1.299866	31	988.347545	14	12.818	5.738	0.129	1	0.566667
3	40.010622	Jokisalo	Birch	9.675957	9.32223	5.2514	43.548687	111.1	1.804322	1.224874	3.029196	1.324863	27	860.818829	12	11.91	5.653	0.14	1	0.576923
2	54.963583	Jokisalo	Spruce	4.74619	5.763	1.5278	29.036193	109.7	0.97618	0.624936	1.601116	0.749923	10	361.187181	1	8.407	5.633	0.271	0.882705	0.777778
3	55.937082	Jokisalo	Birch	21.046371	12.437615	18.9399	175.44185	200.7	3.836025	2.674724	6.510749	2.749716	14	801.044291	4	9.982	5.733	0.194	0.557211	0.769231
2	64.220279	Jokisalo	Spruce	6.515623	7.108231	2.6232	50.865805	134.7	1.291426	0.87491	2.166336	0.949902	28	892.701008	2	15.846	7.141	0.125	1	0.888889
1	49.162954	Jokisalo	Pine	5.915685	13.540231	2.2983	101.8998	137.7	1.237814	0.849912	2.087726	0.899907	27	860.818829	1	16.377	7.663	0.088	1	0.192308

Tree/ha SGINI SMLI

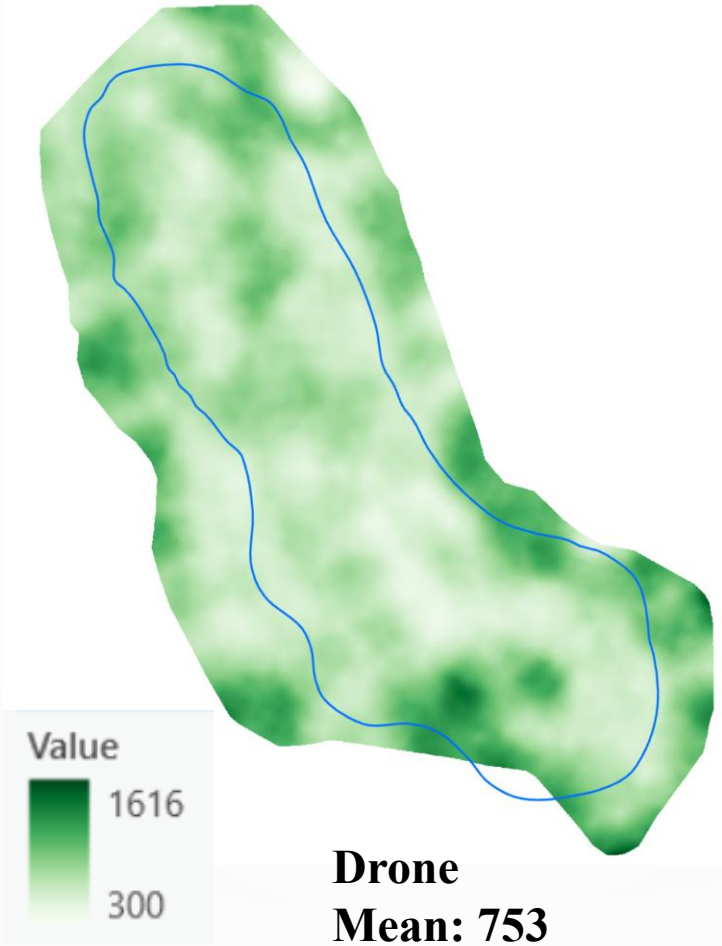
Spatial interpolation

Estimates values at **unsampled locations** using measurements from **known sample points**.

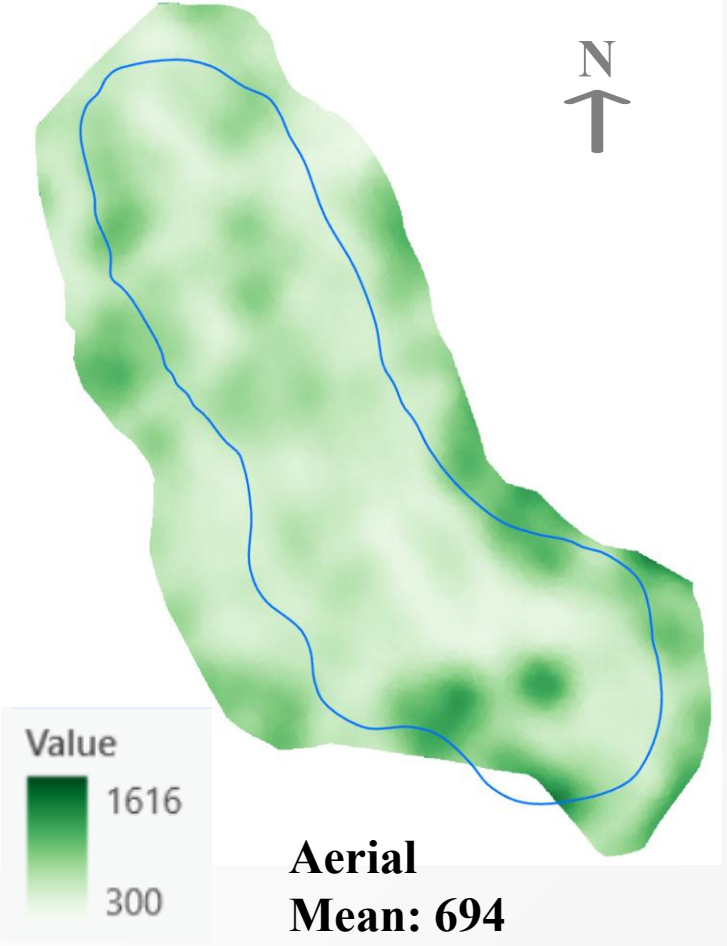
TREE DENSITY (TREE/HA) MAPPING USING SPATIAL INTERPOLATION



Ground truth

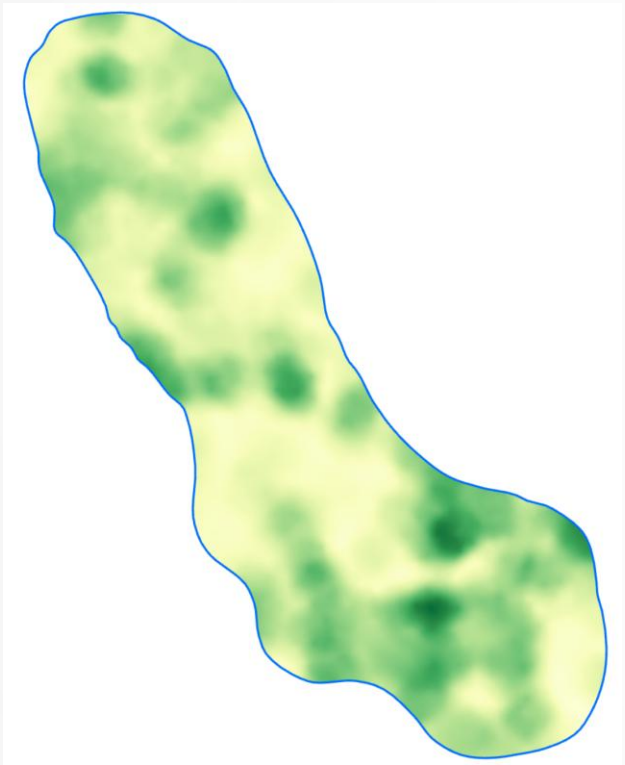


Drone
Mean: 753

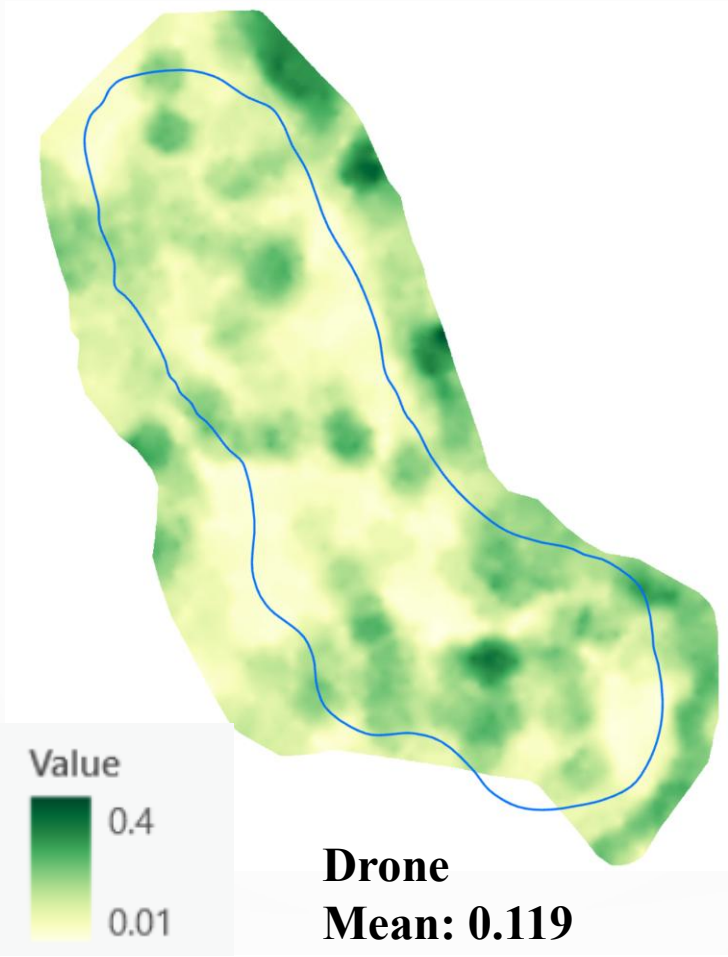


Aerial
Mean: 694

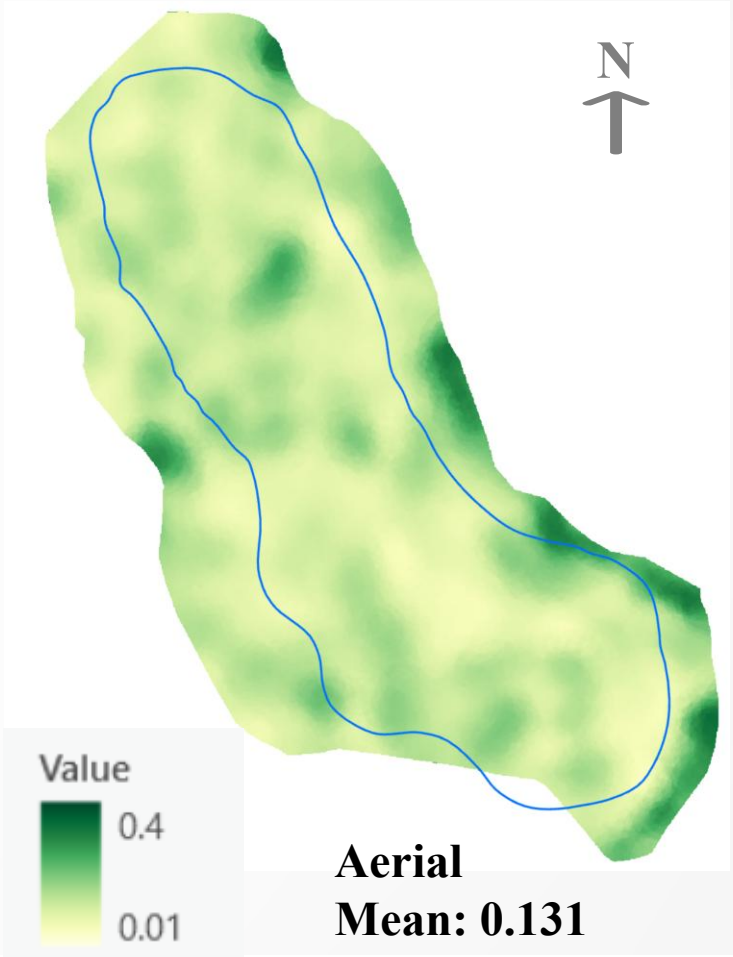
SGINI MAPPING USING SPATIAL INTERPOLATION



Ground truth

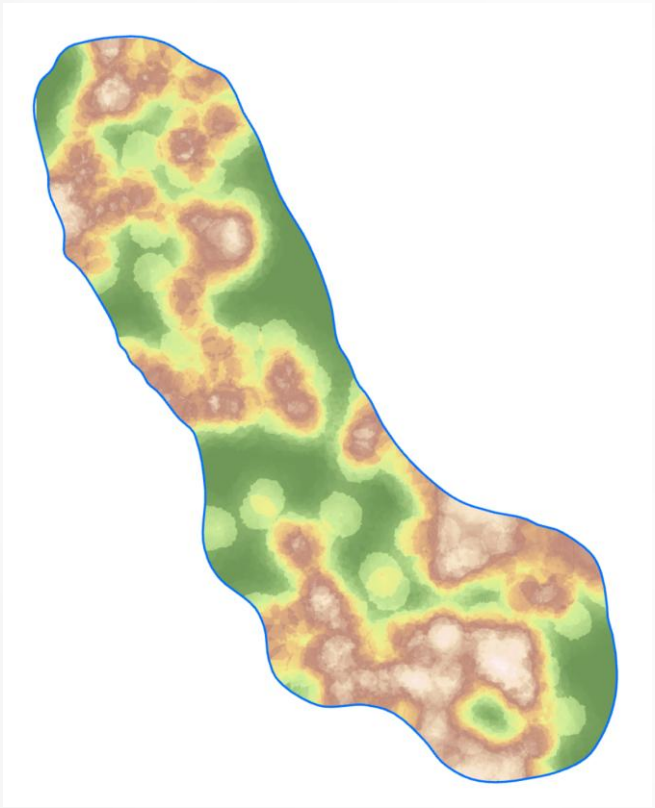


Drone
Mean: 0.119

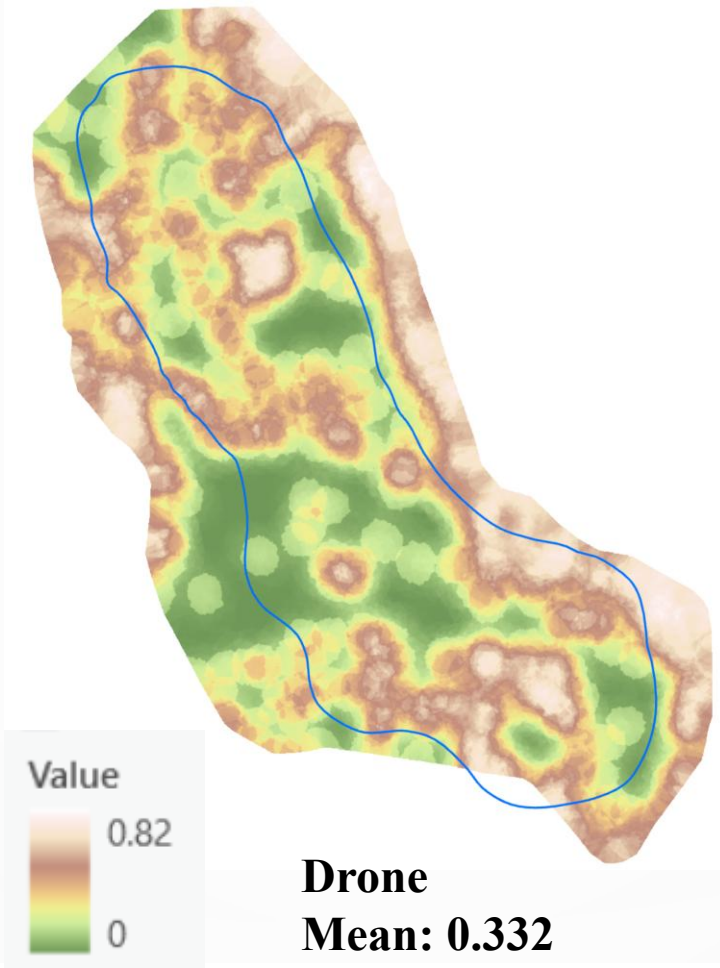


Aerial
Mean: 0.131

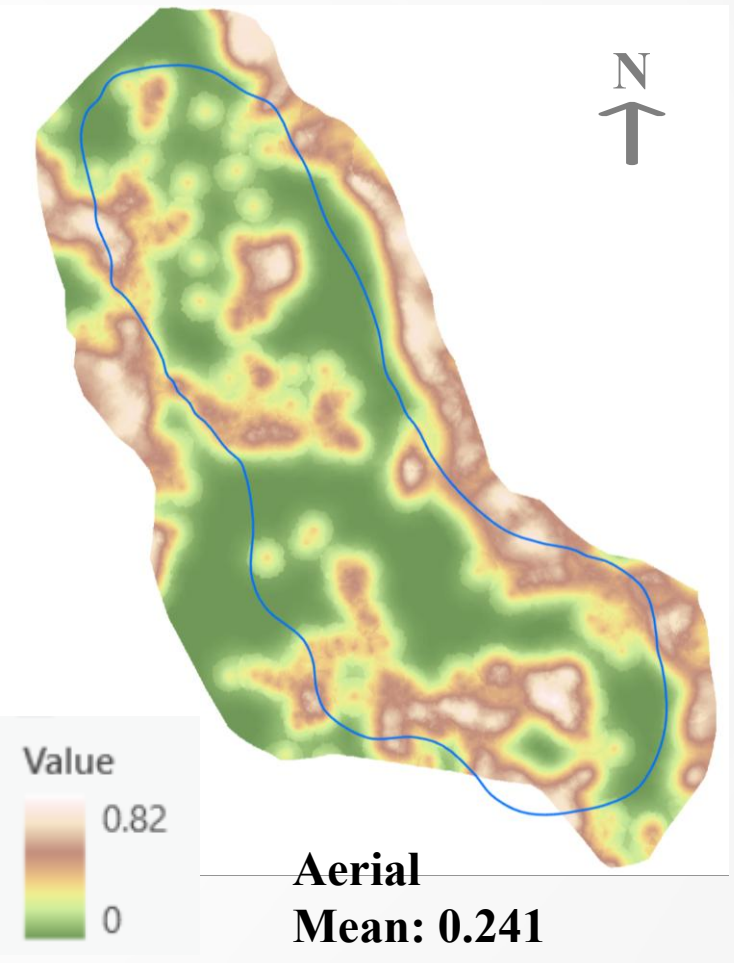
SMLI MAPPING USING SPATIAL INTERPOLATION



Ground truth

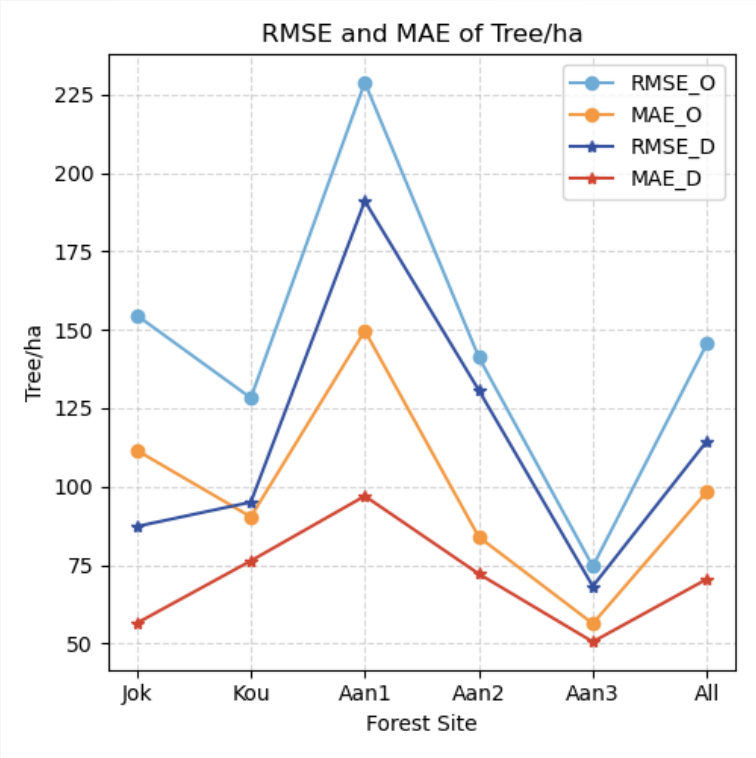


Drone
Mean: 0.332

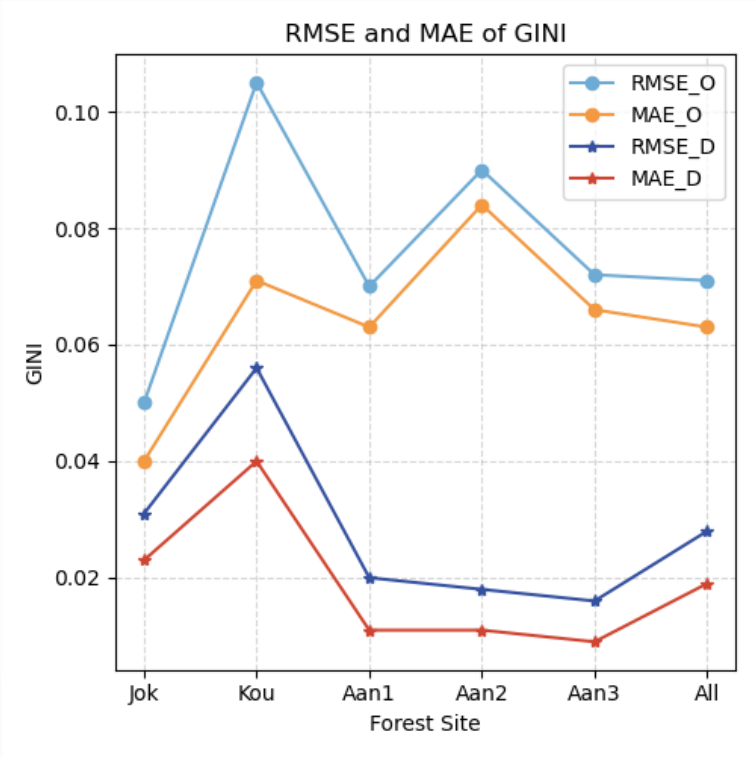


Aerial
Mean: 0.241

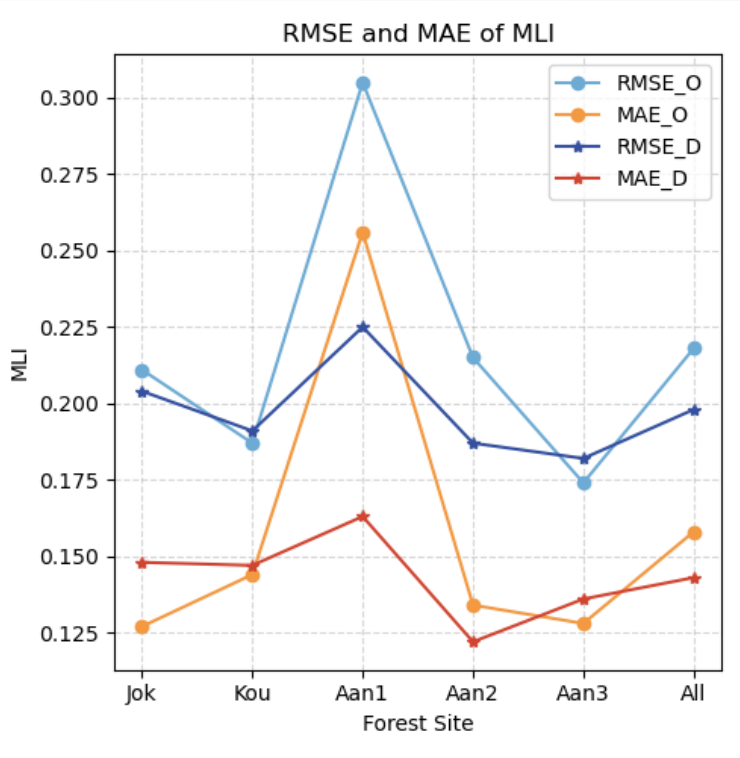
EVALUATION METRICS



Tree density (Tree/ha)



SGINI



SMLI

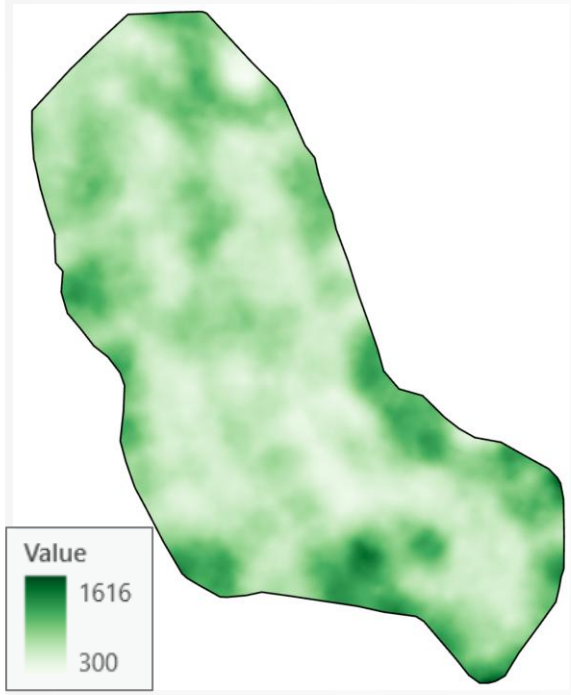
CLUSTERING TO SUPPORT PRACTICAL FOREST OPERATIONS

Saliency detection (regions with high values)

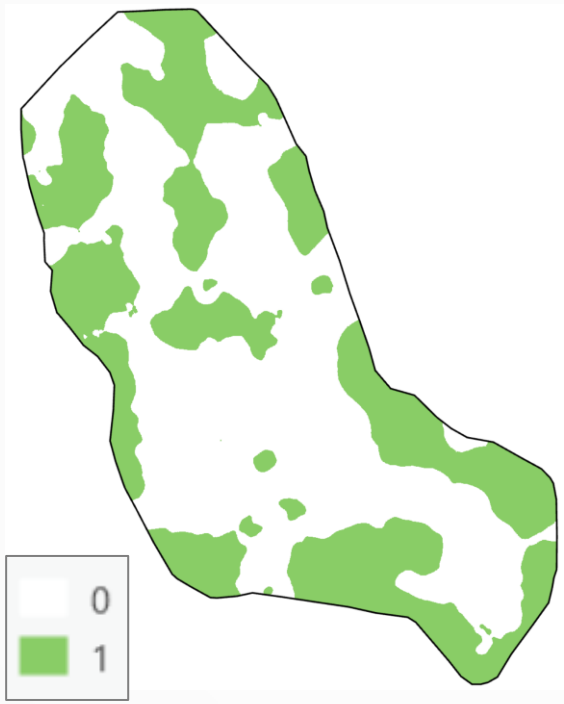
- High tree density
- High tree species/height diversity
- Etc.

Clustering is based on forest structural diversity maps

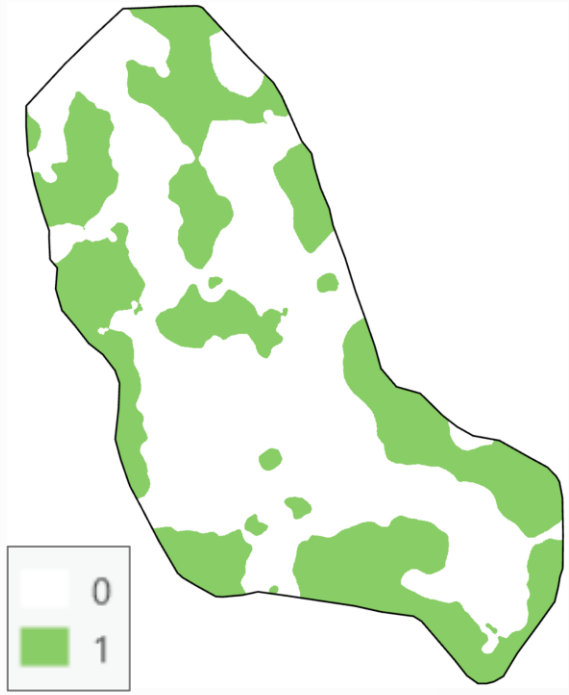
EXTRACT CLUSTERING REGIONS USING THRESHOLDS



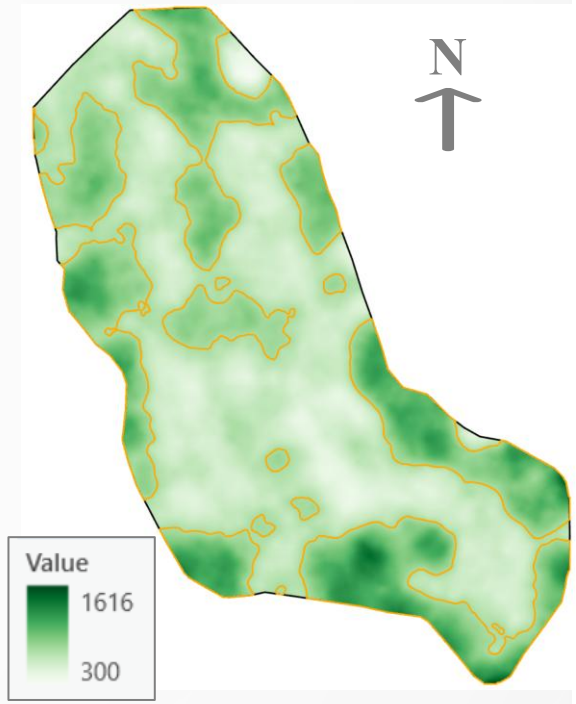
Spatial interpolation result of forest structural indices



Extract regions by threshold (can be defined by users)

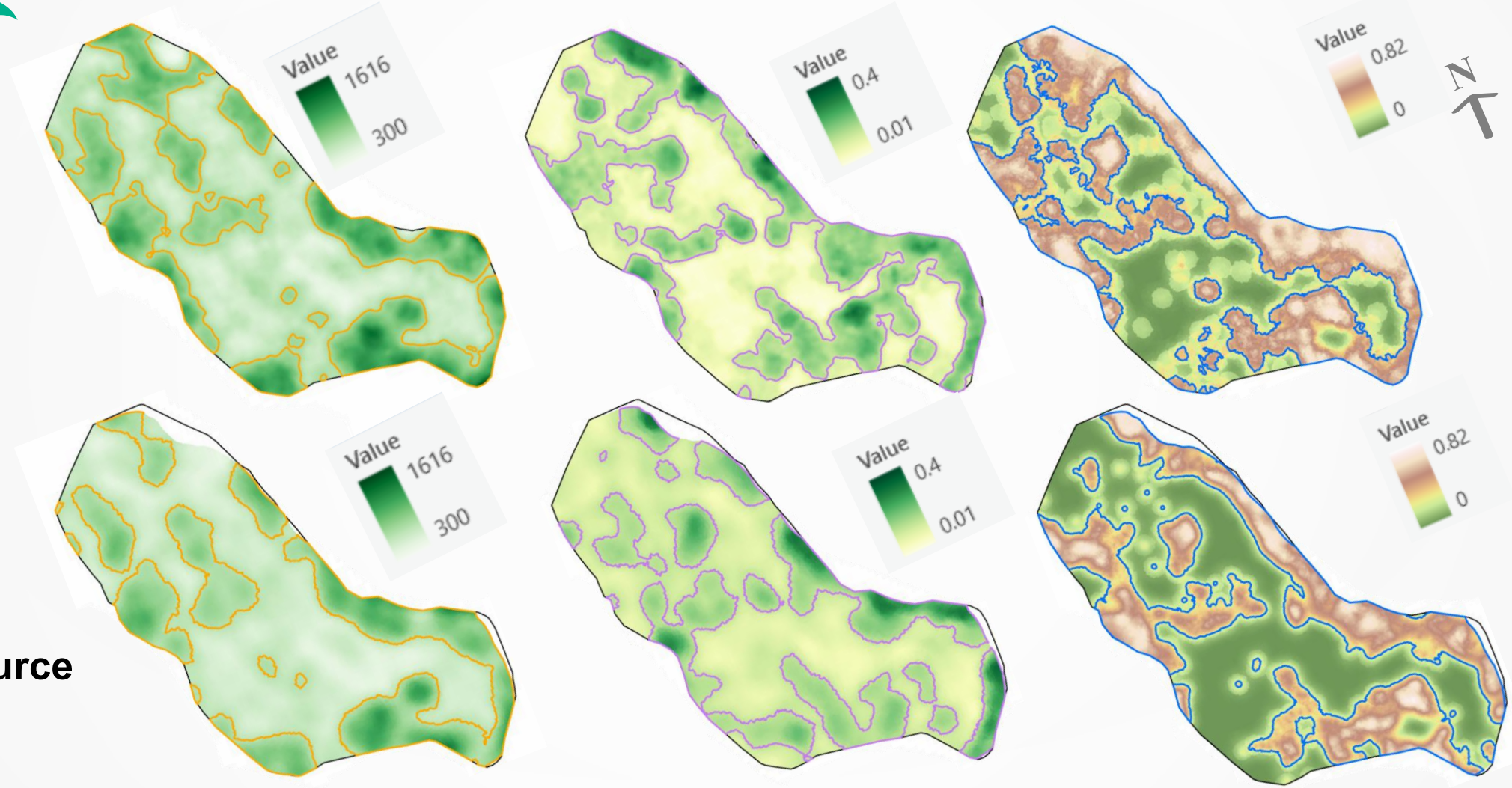


Remove small regions, fill holes, and smooth edges.



Clustering polygons

CLUSTERING: DRONE VS. AERIAL



CONCLUSION

- **Drone data** provide a better representation of forest structure, as their higher spatial resolution enables more accurate individual tree detection.
- **Aerial data** with a 50 cm spatial resolution face greater challenges in individual tree detection, resulting in lower accuracy and poorer detail compared to drone data. However, the overall **pattern of structural diversity** remains consistent between the two datasets.
- **Future work** will focus on integrating **terrestrial laser scanning (TLS)** data to improve the accuracy of aerial-based forest structural diversity analysis.

THANK YOU FOR LISTENING!

Zhu Mao, Omid Abdi, Laamanen Ville, Kivinen Veli-Pekka, Jori Uusitalo